



University of Windsor

COMP - 8977 Internship/Project-2

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Project Proposal

Visual Analytics of Housing Affordability and Housing Market Trends in Windsor Using Canadian Open Data

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1. Introduction

This project will develop a public, web-based visual analytics platform for understanding housing affordability and housing-market conditions in the Windsor Census Metropolitan Area (CMA). The platform will use official Canadian open data, store cleaned information in a relational database, and present the results through interactive Apache Superset dashboards delivered through a React-based website.

The project focuses on a problem that cannot be understood well from one number or one table. Housing affordability depends on the relationship between household income, tenure, shelter costs, rental prices, vacancy, housing supply and the demographic context of the region. Statistics Canada provides detailed census and affordability information, while Canada Mortgage and Housing Corporation (CMHC) provides complementary rental-market and housing-construction data. Bringing these sources together makes it possible to tell a clearer story about who is under housing pressure and how the market is changing.

Before preparing this proposal, the team downloaded and reviewed the proposed files from their original government sources. The initial data audit confirmed that the selected datasets contain the variables required for the planned analysis. The following datasets will be used as the starting point:

Dataset	Source	What we will analyse
Census Profile, 2021 CMAs/CAs - (98-401-X2021002)	Statistics Canada	Windsor CMA population, income, labour, households and housing context
Shelter-cost-to-income ratio by tenure (98-10-0252-01)	Statistics Canada	Affordability by income group, household type, tenure and shelter-cost burden
Rental Market Survey Data Tables - Windsor, 2025	CMHC	Vacancy, average rent, rental universe, turnover, bedroom type and zones
Starts / completions / construction - tables 2026 edition	CMHC	Housing starts, completions and supply activity by dwelling type
Rental Market Survey Data Tables - Canada, 2025 (supporting)	CMHC	Optional Windsor benchmarking against other Canadian centres

The work follows the approach discussed with the project supervisor: begin by understanding the data and the questions it can answer, load validated datasets into a database, analyse them through Apache Superset, and add the React web layer after the analytical core is stable. The emphasis is on trustworthy data integration and visual communication rather than artificial intelligence.

2. Motivation

2.1 Why the problem is interesting

Housing is a strong domain for visual analytics because market conditions and affordability affect many groups differently. In the verified 2021 Census Profile for Windsor CMA, the population was 422,630, up 6.0% from 2016. The same profile shows that 34.8% of tenant households spent 30% or more of income on shelter costs, compared with 10.9% of owner households. These figures immediately suggest questions about which income groups, household types and tenure groups experience the greatest pressure.

The CMHC rental-market data adds a more current market perspective. For Windsor CMA in 2025, CMHC reported a 3.7% purpose-built rental vacancy rate and an average two-bedroom rent of \$1,454. The downloadable Windsor workbook also contains detailed tables by bedroom type, zone, year of construction, turnover and rental universe. Combining this market evidence with Statistics Canada

household and income data creates a more useful story than analysing either source alone.

2.2 Why it is challenging

The project is challenging for several practical reasons:

- **Data integration.** The selected sources use different reporting years and structures: the Census provides a 2021 demographic snapshot, CMHC rental data describes 2025 conditions, and construction data is updated during 2026. These sources should be connected only when geography, time period and definitions make the comparison valid.
- **Data preparation.** Statistics Canada data is supplied in large CSV files and CMHC data is supplied in multi-sheet Excel workbooks. The data must be profiled, reshaped, standardized and documented before dashboard use.
- **Question design.** A chart is only useful when it answers a meaningful question. The team must avoid unsupported claims and reframe questions when the available variables are insufficient.
- **Web accessibility.** The supervisor prefers a free, web-based solution. Power BI is therefore not the primary tool; Apache Superset is preferred because it is open source, database-connected and suitable for online dashboards.
- **Communication.** The final dashboards must make sense to non-technical users and should present connected stories instead of a collection of unrelated visualizations.

2.3 Who benefits

- Residents and students who want a clearer view of rent, vacancy, affordability and housing supply in Windsor.
- Community organizations and local decision makers who need an accessible summary of housing conditions and demographic context.
- Researchers and future students who can reuse the documented datasets, SQL views, ETL scripts and dashboards.
- The project team, which gains practical experience with an end-to-end data engineering, database, analytics and web-delivery workflow.

3. Background Study

3.1 Open government data in Canada

Statistics Canada and CMHC publish authoritative public data that can be downloaded directly from the original sources. Using the original agencies provides traceability, documentation and clear definitions. The project will preserve the raw downloads separately from processed data so that each dashboard metric can be traced back to its source.

3.2 Selected datasets

Census Profile, 2021 - Statistics Canada. The selected CMA/CA profile file contains 2,631 characteristics for Windsor CMA, including population, household income, labour-force indicators, housing tenure, shelter costs and other demographic variables. It provides the broad demographic and economic context for the project .

Shelter-cost-to-income ratio by tenure -Statistics Canada. Table 98-10-0252-01 is the project's main affordability table. The downloaded Windsor CMA file contains 2,389 rows and dimensions for household total income, household type, shelter-cost-to-income ratio and tenure. It allows the team to analyse housing-cost burden for renters and owners across household and income groups .

Rental Market Survey Data Tables - CMHC. The Windsor 2025 workbook contains 29 worksheets. It includes vacancy-rate estimates, average rents, rental-universe counts, turnover, bedroom types, zones, year of construction and additional affordability-related breakdowns. CMHC also publishes historical Windsor workbooks, which can be incorporated later if a longer time series is needed.

Housing starts and completions - CMHC. The selected construction workbook includes CMA-level tables for Windsor and provides starts and completions by dwelling type. This source supports the housing-supply story and can be refreshed with later editions without changing the overall architecture.

3.3 Related analytical approach

The supervisor emphasized a story-first approach: choose a domain, understand the available variables, ask valid questions, and add a second dataset when one source cannot support the full question. For example, a question about which groups face greater housing affordability pressure requires household income and tenure information in addition to market-level rent data. The proposed project therefore uses several related datasets to support connected analytical stories rather than forcing all records into one large table.

The project will distinguish description from interpretation. Superset will be used to display KPIs, trends, comparisons and filters, while the written narrative will explain what the evidence supports. The system is not planned as an LLM or chatbot; the 'answers' are the findings derived from the data, SQL calculations and interactive visualizations.

3.4 Tools and technologies reviewed

Tool	Type	Decision
Apache Superset	Web BI / dashboard platform	Selected - primary analytics layer; free, web-based, SQL-connected and embeddable
Power BI	Desktop / cloud BI	Not selected - supervisor requested a free web-based approach; publishing can require licensing
PostgreSQL	Relational database	Selected - central store for cleaned analytical data and SQL views
Python (pandas, SQLAlchemy)	Data processing	Selected - profiling, cleaning, reshaping, loading and repeatable refresh scripts
React	Front-end web library	Selected - narrative website and linked/embedded Superset dashboards
Docker	Containerisation	Planned - reproducible local setup for Superset and PostgreSQL

4. Project Goals

4.1 Vision

Give anyone with a web browser a clear, trustworthy and interactive picture of housing affordability and housing-market conditions in Windsor CMA, using only official Canadian open data and a transparent analytical process.

4.2 General goals

1. Tell a small number of connected housing stories that a non-technical visitor can follow from question to evidence to conclusion.
2. Show which household, income and tenure groups experience the greatest shelter-cost burden.
3. Present rental-market indicators such as rent, vacancy, rental stock and turnover in an understandable form.
4. Describe current housing-supply activity through starts and completions and place it beside Windsor demographic context.
5. Keep every figure traceable to its official source and avoid invalid joins across incompatible years or geographies.
6. Produce a reusable, refreshable project that can accept newer government releases without redesigning the dashboards.

4.3 Guiding questions (first draft)

These questions will guide the dashboard stories and may be refined after the full data audit:

Story 1 Who faces the greatest housing affordability pressure?

- How does shelter-cost burden vary by income group and household type?
- How different is the burden for renters and owners?

Story 2 What is happening in Windsor's rental market?

- How do average rents, vacancy, rental stock and turnover vary by unit type and zone?
- Which parts of the market appear tighter or more affordable?

Story 3 What does housing supply look like?

- How many housing units are being started and completed, and which dwelling types contribute most?
- How can supply activity be interpreted alongside Windsor population and household context?

Story 4 How does Windsor compare?

- Where compatible data exists, how does Windsor compare with selected Canadian CMAs on rental indicators?

4.4 Expected outputs

- A public React website introducing the project, methodology, sources and analytical stories.
- Interactive Apache Superset dashboards with KPIs, filters and comparisons.
- A documented PostgreSQL database containing cleaned and structured datasets.
- Repeatable Python ETL scripts for loading and refreshing source data.
- A GitHub repository, final report, proposal/final presentations and Demo Day demonstration.

5. Roadmap of the Project

The project runs for 11 weeks, from September 28 to December 9, 2026. The work is divided into five phases: (1) understand and validate the data, (2) build the data pipeline and database, (3) analyse and create Superset dashboards, (4) integrate the web layer and deployment, and (5) test, document and present. The weekly dates and team-role structure follow the team's existing proposal plan.

5.1 Team roles

Team member	Lead role	Main responsibilities
Dhritiben Patel	Project Manager and Narrative Lead	Sprint tracking, meeting notes, supervisor communication, data storytelling, report and presentation coordination
Romit Patel	Data Engineering Lead	Dataset acquisition, profiling, Python ETL, PostgreSQL schema, data validation and refresh pipeline
Pavan Rella	Analytics and Visualisation Lead	Apache Superset setup, analytical SQL, KPIs, charts, dashboards and filters
Charmiben Patel	Front-end and Deployment Lead	React website, Superset integration, hosting research, deployment and GitHub coordination

5.2 Weekly deliverables

Week	Dates	Phase	Deliverables
1	Sep 28 – Oct 4	Initiation	Supervisor meeting; housing domain explored; source shortlist; architecture and team roles discussed

Week	Dates	Phase	Deliverables
2	Oct 5 – Oct 11	Initiation	Proposal submitted Oct 7; proposal presentation Oct 8; core StatsCan and CMHC files downloaded and validated
3	Oct 12 – Oct 18	Data understanding	Full data audit: fields, years, geographies, missing/suppressed values; final questions and data dictionary
4	Oct 19 – Oct 25	ETL / database	Python cleaning pipeline; PostgreSQL schema and ER model; first dataset loaded
5	Oct 26 – Nov 1	Database / Superset	All core datasets cleaned and loaded; analytical SQL views; Superset installed and connected
6	Nov 2 – Nov 8	Analysis	Exploratory SQL; first affordability and rental-market charts; supervisor progress review
7	Nov 9 – Nov 15	Dashboards	Affordability and rental dashboards complete; KPI definitions and native filters validated
8	Nov 16 – Nov 22	Dashboards / web	Housing-supply and comparison story; React website skeleton; Superset linking/embedding
9	Nov 23 – Nov 29	Deployment	Website and dashboards deployed; refresh process documented; narrative text finalized
10	Nov 30 – Dec 6	Testing / documentation	Data and user testing; bug fixes; final report draft; README and architecture documentation
11	Dec 7 – Dec 9	Closing	Final report, presentation rehearsal, Demo Day demonstration and project handover

5.3 Weekly responsibilities of each team member

Week	Dhritiben Patel	Romit Patel	Pavan Rella	Charmiben Patel
1	Set up Jira; record supervisor requirements	Research/verify StatsCan and CMHC data	Review Superset and possible analytics stories	Create repo structure; review React/hosting approach
2	Write intro, motivation and goals; coordinate proposal	Validate downloaded files; document dataset coverage	Write background/tool review; draft dashboard questions	Write resources/roadmap; prepare proposal slides
3	Define audience and narrative flow	Profile all files; document fields and data quality	Map questions to measures and visualizations	Compare hosting/deployment options
4	Track sprint and review schema	Build ETL and PostgreSQL schema	Learn Superset; test SQL datasets/charts	Prepare deployment environment and React plan
5	Maintain data dictionary and decisions	Clean/load core datasets; validate geography/time rules	Install/connect Superset; create analytical views	Support Superset environment and repository configuration
6	Prepare progress review and story outline	Fix data-quality issues and join logic	Exploratory SQL; first affordability/rental charts	Design React wireframes/navigation
7	Write narrative for first stories	Create reusable SQL views and metrics	Complete affordability/rental dashboards and filters	Build React skeleton and source/method pages
8	Review dashboard clarity and evidence	Prepare supply/comparison datasets	Build supply/comparison dashboard and KPIs	Link or embed Superset in React
9	Finalize story text and project status	Implement/document refresh process	Tune filters, performance and dashboard consistency	Deploy/test website on desktop/mobile
10	Coordinate user testing and report assembly	Write pipeline/database report sections	Write analysis/findings sections	Write implementation/deployment section; README
11	Finalize report and lead presentation	Present data pipeline and validation	Present dashboards and findings	Run live website/demo deployment

6. Resources

6.1 Required skills

- Databases: relational modelling, SQL, PostgreSQL, analytical views and data validation.
- Data processing: Python, pandas and SQLAlchemy for reading CSV/XLSX, cleaning, reshaping and loading data.
- Visualisation: Apache Superset charts, KPIs, dashboard filters and principles of clear visual design.
- Web development: React, HTML, CSS and JavaScript; linking or embedding Superset dashboards.
- Deployment: Docker, environment configuration and free/low-cost web hosting.
- Domain and communication: understanding housing indicators, data storytelling, teamwork and technical writing.

6.2 Project management tools

Tool	Purpose
Jira	Sprint planning, backlog, task assignment, progress tracking and meeting decisions
GitHub	Version control, code review, issue tracking and public portfolio repository
Microsoft Teams	Day-to-day team communication and quick coordination

6.3 Technology stack

Layer	Technology	Role in the project
Data sources	Statistics Canada; CMHC	Official Census, affordability, rental-market and construction datasets
Data processing	Python (pandas, SQLAlchemy)	Profile, clean, reshape, validate and load source data
Database	PostgreSQL	Central store for cleaned tables, SQL views and analytical metrics
Analytics	Apache Superset	SQL exploration, KPIs, filters, charts and interactive dashboards
Front end	React	Public story pages, methodology/source pages and Superset integration
Containerisation	Docker	Consistent local setup of Superset and PostgreSQL

6.4 Hosting approach

The project will prioritize a free or sustainable web-based deployment rather than a desktop-only BI solution. Hosting will be evaluated based on public access, cost, resource limits and the ability to run PostgreSQL, Superset and React. A final hosting choice will be made after the core local pipeline is stable.

6.5 Data refresh

The first version will use validated batch downloads. The ETL scripts will then be designed so the same workflow can be re-run when official agencies publish updated files. Where source structure

remains stable, refresh can be scheduled daily or weekly; if a schema changes, validation will stop the load rather than silently corrupt the dashboards. The project does not require second-by-second real-time data.

7. References

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